Project Description:

The Farm Management System is a Salesforce-based application designed to simplify and organize farm-related activities and information. The system provides a centralized platform for managing farmers, crops, buyers, sales, and other important farm records.

The application helps reduce manual record keeping and improves the efficiency of managing agricultural information. Salesforce objects, fields, relationships, validation rules, flows, and security configurations are used to maintain and process farm-related data.

The system allows authorized users to create, view, update, and manage farm records in an organized manner. It also supports better coordination between different users or departments involved in farm management.

Scenarios

Scenario 1: A farmer's information is entered into the system along with relevant farm and crop details. The records can be stored and accessed through Salesforce.

Scenario 2: A buyer requires information about available farm products. The user can maintain buyer and product-related records and manage the required information systematically.

Scenario 3: The Sales department manages buyer and sales-related information. The system helps maintain these records and provides organized access to the required information.

Technical ArchitUser

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Salesforce User Interface

↓

Salesforce Objects & Fields

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Relationships / Validation Rules

↓

Flows & Automation

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Farm Management Data Architecture:

Description:

The Farm Management System uses Salesforce as its cloud-based application platform. The system consists of Salesforce objects, fields, relationships, automation, security configurations, and user interfaces.

Users interact with the Salesforce application through the user interface. The entered information is stored in Salesforce and managed using the configured objects and relationships. Salesforce automation features can be used to perform required actions and maintain consistency in the records.

Pre-requisites

- **Basic knowledge of Salesforce**
- **Salesforce Developer/Trailhead environment**
- **Understanding of Salesforce Objects and Fields**
- **Knowledge of Relationships between objects**
- **Basic knowledge of Salesforce Flow**
- **Understanding of Validation Rules**
- **Basic knowledge of Salesforce User and Permission management**
- **Basic understanding of CRM and farm management processes**

Project Workflow



Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1: Configure the Salesforce environment and project access.**
- **Activity 1.2: Identify and select the Salesforce features required for the Farm Management System.**
- **Activity 1.3: Define the architecture of the application, including objects, fields, relationships, automation, and users.**
- **Activity 1.4: Set up the Salesforce development environment and configure the required project components.**

Milestone 2: Core Functionalities Development

- **Activity 2.1: Develop the core Farm Management functionalities such as farmer, crop, buyer, product, and sales management.**
- **Activity 2.2: Configure Salesforce objects, fields, relationships, validation rules, and automation required for the system.**

MilesMilestone 3: App.py Development

- **Activity 3.1: Configure the main application components and Salesforce functionalities required to manage Farm Management System records.**

Milestone 4: Frontend Development

- **Activity 4.1: Design and configure the Salesforce user interface using appropriate Salesforce pages and components.**
- **Activity 4.2: Configure dynamic record views, forms, related lists, and user interactions according to the project requirements.**

Milestone 5: Deployment

- **Activity 5.1: Prepare the Salesforce application and verify all configurations, objects, fields, and automation.**
- **Activity 5.2: Test the application in the Salesforce environment and make it accessible to authorized users.**

Milestone 6: Conclusion

The Farm Management System is completed by integrating the configured Salesforce components and verifying the major functionalities.



Milestone 1: Model Selection and Architecture

In this milestone, the required Salesforce features are identified and the overall architecture of the Farm Management System is designed.

The system architecture focuses on managing farm-related information through Salesforce objects and relationships. The required fields, users, permissions, automation, and data flow are identified before implementation.

Activity 1.1: Generate a Groq API Key

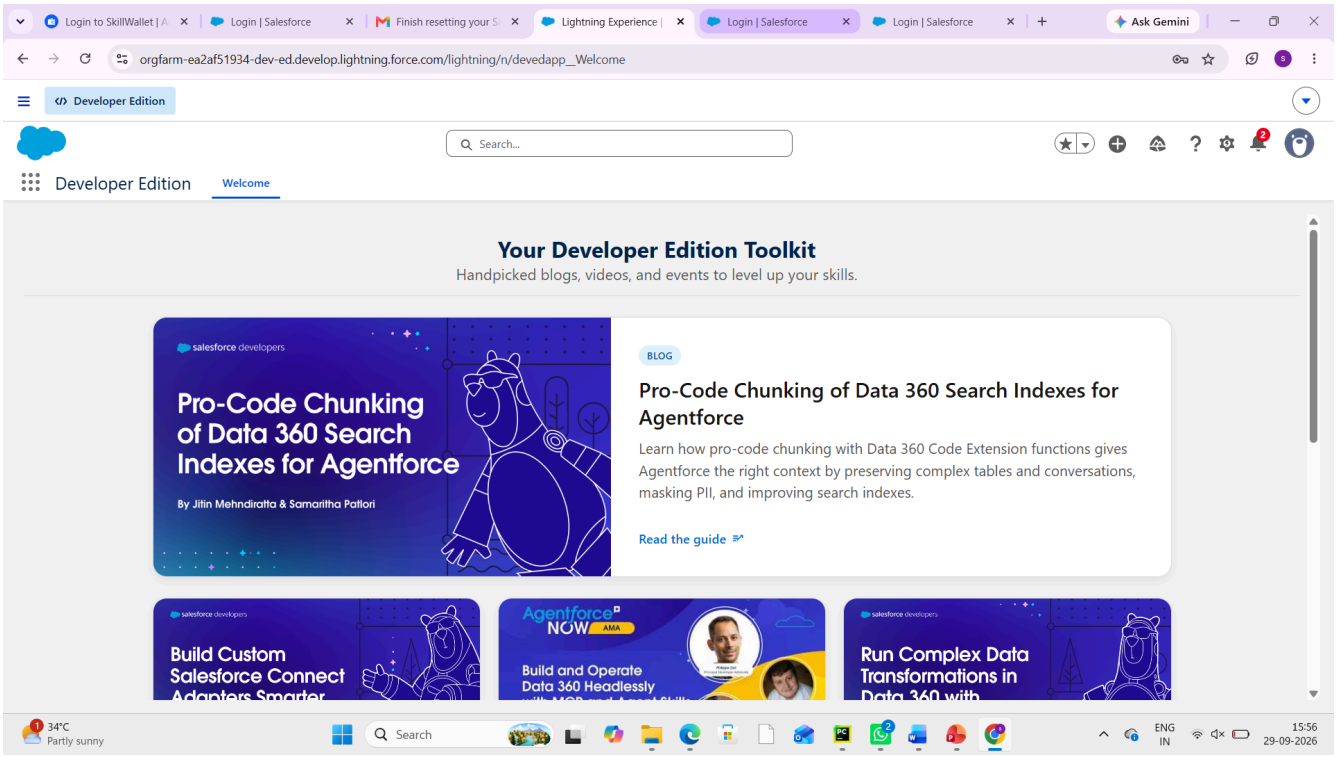
For the Farm Management System, the project does not require a Groq API key. The application is implemented using Salesforce configuration and metadata.

The Salesforce environment provides the required platform for developing and accessing the Farm Management System.

Activity 1.2: Research and Select the Appropriate Generative AI Model

The Farm Management System does not depend on a Generative AI model as its primary functionality.

Instead, the project focuses on Salesforce features such as:



Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code(files and folders)	Yes(Salesforce configuration and metadata-farm Management data)
2	README / Setup Guide	Yes
3	requirements.txt / package.json / dependency file	NA
4	Database schema / seed files	YES
5	Environment configuration file	NA
6	Deployed application URL	NA
7	APK / Executable binary	YES

- Custom Objects

- Custom Fields
- Relationships
- Validation Rules
- Salesforce Flow
- User Permissions

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7	APK / Executable binary	YES

- Record Management

These features are selected based on the requirements of the Farm Management System.





• **Step 3 — Create a New API Key:**

Activity 1.3: Define the Architecture of the Application

The architecture defines how users interact with the Farm Management System and how information is stored and processed.

The major components include:

- **Salesforce User Interface**

- **Farm-related Objects**
- **Custom Fields**
- **Object Relationships**
- **Validation Rules**
- **Salesforce Flow**
- **User Permissions**
- **Farm Management Records**

The architecture provides a centralized system for maintaining farm-related information.

- . **Activity 1.4: Set Up the Development Environment**

The Salesforce environment is prepared for project development.

The required steps include:

1. Access the Salesforce development environment.
2. Create/configure the required Salesforce objects.
3. Create the required fields.
4. Configure relationships between objects.
5. Add required sample records.
6. Configure validation rules.
7. Configure Salesforce Flow and automation.
8. Configure user access and permissions.
9. Test the application.

Milestone 2: Core Functionalities Development

Activity 2.1: Develop Core Content Generation Features

For the Farm Management System, the core functionalities are related to **farm and sales management** rather than content generation.

The system can include:

- Farmer Management
- Crop Management
- Product Management
- Buyer Management
- Sales Management
- Farm Record Management
- Data Management

These functionalities help users maintain farm-related information in one centralized Salesforce application.

Activity 2.2: Implement the Flask Backend

The Farm Management System does not use Flask as its backend

Salesforce itself provides the application platform, database, security, and automation environment. Therefore, the required functionality is implemented using Salesforce configuration, metadata, objects, relationships, and automation.

Milestone 3: App.py Development

This milestone is adapted to the Salesforce implementation of the Farm Management System.

Activity 3.1: Writing the Main Application Logic in app.py

The Farm Management System does not use an `app.py` file because it is developed using Salesforce.

The corresponding application logic is handled through Salesforce configuration and automation, including:

- Salesforce Objects
- Fields
- Relationships
- Validation Rules
- Flows
- Permission Sets
- Page Layouts
- Lightning Record Pages

These components together provide the required functionality of the application.

Define Prompt Engineering Helpers

The Farm Management System does not require prompt engineering because it is not dependent on an AI content-generation model.

Instead, the project uses Salesforce configuration and automation to manage farm-related data.

The main configuration components include:

- Object configuration
- Field configuration
- Relationship configuration
- Validation rules
- Flow automation
- User permissions

Integrate the Groq AI Responses in Each Function

The Farm Management System does not use Groq AI responses.

Instead, user-entered farm-related information is stored and processed within Salesforce. The configured objects, fields, relationships, and automation support the management of the records.

Milestone 4: Frontend Development

This milestone focuses on configuring a user-friendly Salesforce interface for the Farm Management System.

The Salesforce interface allows authorized users to access and manage farm-related records.

Activity 4.1: Designing and Developing the User Interface

The Salesforce interface is configured to provide access to the required Farm Management System modules.

The interface can contain:

- Farmer records
- Crop records
- Product records
- Buyer records
- Sales records
- Related information
- Record creation and editing options

The appropriate page layouts and Lightning pages are configured according to the project requirements.

Activity 4.2: Creating Dynamic Content Rendering with JavaScript

The Farm Management System does not require custom JavaScript for its basic Salesforce implementation.

Salesforce automatically displays and updates records through its standard interface and configured components.

When users create or update records, the changes are reflected in the corresponding Salesforce records and related lists.

Milestone 5: Deployment

The Farm Management System is deployed and accessed through the Salesforce environment.

Activity 5.1: Preparing the Application for Local Deployment

Since the project is developed using Salesforce, traditional local deployment using Flask or Python is not required.

The project is prepared by verifying:

- Salesforce objects
- Fields
- Relationships
- Sample data
- Validation rules
- Flows
- Permissions
- Page layouts
- Lightning pages

Activity 5.2: Local Testing and Verification

The configured Salesforce application is tested to ensure that the major functionalities work correctly.

Testing includes:

1. Creating farm-related records.
2. Updating existing records.
3. Checking object relationships.
4. Testing validation rules.
5. Testing Salesforce Flow automation.
6. Checking user access.
7. Verifying buyer and sales information.
8. Checking that the records are stored correctly.

Activity 5.3: Public Deployment via Ngrok

The Farm Management System does not require Ngrok because the application is hosted on the Salesforce cloud platform.

Authorized users can access the system through the Salesforce environment using the appropriate login credentials and permissions.

Exploring the Website's Web Pages

Home / Generator Page:

For the Farm Management System, the corresponding main Salesforce application page provides access to the major farm management modules.

The main interface allows users to navigate to the required records and perform permitted operations.

The application can provide access to:

- Farmer Management
- Crop Management
- Product Management
- Buyer Management
- Sales Management

Input Section

The Farm Management System provides record-entry forms through Salesforce.

Users can enter the required information into the configured fields.

Examples include:

- Farmer details
- Crop details
- Product details

- Buyer details
- Sales details

Validation rules help ensure that the required information is entered correctly.

Results Section

After records are created or updated, the information is displayed through Salesforce record pages and related lists.

Users can view the stored farm-related information and its associated records.

The system provides organized access to the information based on the user's permissions.

Conclusion

The **Farm Management System** is a Salesforce-based application developed to organize and manage farm-related information efficiently. The system provides a centralized platform for maintaining farmer, crop, product, buyer, and sales information.

By using Salesforce objects, fields, relationships, validation rules, automation, and user permissions, the system reduces manual record management and improves the organization of farm data.

The project demonstrates how Salesforce can be used to develop a structured Farm Management System with centralized data management and controlled user access.

The system can be further enhanced in the future by adding additional automation, reports, dashboards, notifications, advanced analytics, and other farm-management functionalities.
